physics.’ And I always asked, ‘what did he do?’ ‘Oh, he did something, you don’t have to worry about it, because quantum mechanics works anyhow.’ Nobody at CERN could explain what the Bell inequalities are.” But one day in the fall of 1980, while Bertlmann was visiting Vienna for several weeks, he was suddenly confronted by Bell’s theorem in an unexpectedly personal way. A colleague of Bertlmann’s came running down to his office brandishing a new paper by Bell. “He just came in waving this [paper],” Bertlmann recalled. “And he said, “Reinhold, look what I have!Now you are famous!”

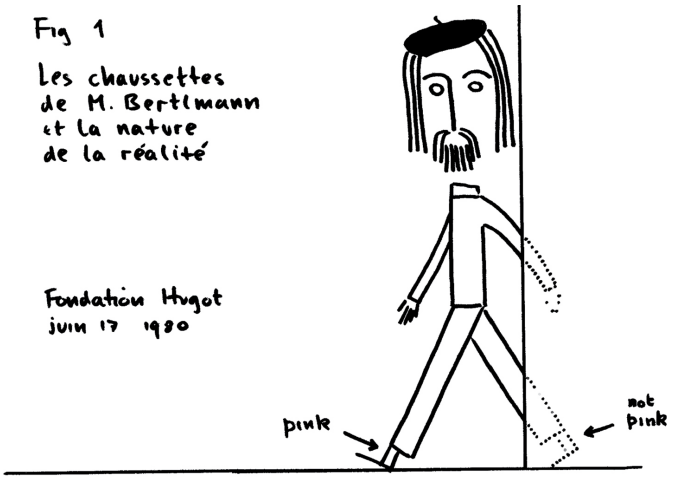


Figure 10.1. John Bell’s cartoon of Bertlmann’s socks, 1980.

Bertlmann, astonished, read and reread the title of the paper: “Bertlmann’s Socks and the Nature of Reality.” The paper even came with a small cartoon, drawn by Bell himself (Figure 10.1).

“The philosopher on the street, who has not suffered a course in quantum mechanics, is quite unimpressed by Einstein-Podolsky-Rosen correlations,” Bell wrote. “He can point to many examples of similar correlations in everyday life. The case of Bertlmann’s socks is often cited. Dr. Bertlmann likes to wear two socks of different colors. Which color he will have on a given foot on a given day is quite unpredictable. But when you see that the first sock is pink you can be already sure that the second sock will not be pink.… There is no accounting for tastes, but apart from that, there is no mystery here. And is not the EPR business just the same?” Bell briefly outlined the Copenhagen interpretation and its history,